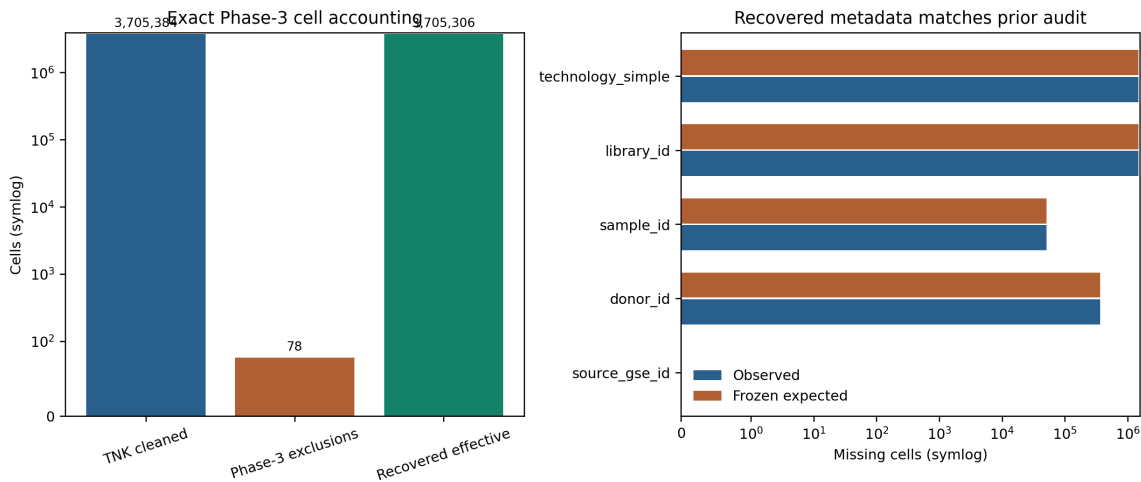


gdTAI V4.2 current-atlas recovery preflight

Decision

PASS_REVIEW_REQUIRED. The SSD-resident `integrated.h5ad` used by the original V4.2 preflight is no longer present. No exact file copy was found in the project or databank search. The canonical NFS `TNK_cleaned.h5ad` can reconstruct the exact raw-count input needed by the V4.2 sidecar without using old scVI coordinates or Phase-4 scores.

- Raw cleaned source: **3,705,384 cells x 27,413 genes.**
- Saved Phase-3 exclusions intersecting this source: **78.**
- Effective recovery: **3,705,306 cells**, exactly matching the frozen preflight.
- Primary dual-annotation NK anchors recovered: **21,054/21,054.**
- Required metadata audit: **all five columns match the previous missing and unique counts exactly.**
- Project-data integration, scVI fitting, clustering, pseudo-labeling, and classifier fitting performed: **no.**



Recovery equivalence

Why recovery is valid for this task

V4.2 integration consumes raw counts, cell identifiers, source, donor, sample, library, and technology. It does not consume the missing object's old scVI latent representation, Leiden labels, UMAP, Phase-4 scores, or cell-type annotation. `TNK_cleaned.h5ad` is the canonical pre-Phase-3 raw-count milestone. Applying the original Phase-3 exclusion manifest removes exactly the 78 cells that distinguish its 3,705,384 rows from the frozen 3,705,306-cell atlas input. The original harmonized metadata files reproduce the five required metadata columns and their prior audit counts.

The runner now permits this recovery only when the original integrated path is absent. It requires the exact source, exclusion-manifest, and metadata hashes; the expected raw/effective dimensions; exact exclusion intersection; and this reviewed recovery contract. If the original integrated H5AD reappears, recovery fails closed rather than choosing between two inputs.

Metadata equivalence

column	observed_missing	expected_missing	observed_unique	expected_unique	exact_match
source_gse_id	0	0	25	25	True
donor_id	366518	366518	687	687	True
sample_id	51007	51007	767	767	True
library_id	1483150	1483150	412	412	True
technology_simple	1483150	1483150	3	3	True

Exclusion accounting

phase3_sanitization_reason	size
extreme_library_size	78

Checks

check	status	detail
original_integrated_h5ad_absent	PASS	/home/tanlikai/databank/publicdata/tools/output_geo_tcell_research/high_speed_temp/Integrated_dataset/integrated.h5ad
recovery_source_sha256	PASS	04033762f9b1380227f0dabbdbdcd7f0fbc4153e734390b30ef64824e3b181ad
phase3_exclusion_manifest_sha256	PASS	dcc9d7865e0a1712e80b8eb621ab93e5bfc9296ec04b1563195f0aa352f1f097
metadata_source_sha256::0	PASS	7fa54bb8ea2a147b84beddcae1f016c4bbb1af4faa9f66c1166e176831eaddac
metadata_source_sha256::1	PASS	e968424576e5859b5d3ecfedf8ef4b772cba71bbdde249a5cc10800f85ea197b
recovery_raw_shape	PASS	(3705384, 27413)
recovery_sparse_csr	PASS	csr_matrix
recovery_unique_axes	PASS	obs_duplicates=0; var_duplicates=0
recovery_raw_count_sample	PASS	{"finite": true, "integer_like": true, "maximum": 1866.0, "minimum": 1.0, "n_sampled": 150000, "nonnegative": true}
phase3_exclusion_intersection	PASS	78 rows
effective_cell_count	PASS	3705306 cells
metadata_equivalence::source_gse_id	PASS	missing=0; unique=25
metadata_equivalence::donor_id	PASS	missing=366518; unique=687
metadata_equivalence::sample_id	PASS	missing=51007; unique=767
metadata_equivalence::library_id	PASS	missing=1483150; unique=412
metadata_equivalence::technology_simple	PASS	missing=1483150; unique=3
primary_nk_anchor_recovery	PASS	21054/21054
primary_nk_anchor_source_count	PASS	2 sources
locked_cohorts_not_opened	PASS	recovery audit reads only current-atlas precursor, exclusion manifest, metadata, and label manifest
all_recovery_inputs_unchanged	PASS	4/4 size/mtime pairs unchanged

Remaining supervision gates

The previous execution approval is intentionally invalidated because the execution config, runner, core, and physical input contract changed. Explicit approval of `RECOVERY_EXECUTION_APPROVAL.json` authorizes only development-data staging, A100 scVI, repeated clustering, and pseudo-NK QC. Classifier fitting, threshold selection, promotion, release fitting, and atlas inference remain separate unapproved actions.

Exact artifacts

- Recovery contract: `configs/models/gdtai/v4_2_integration_execution.json`
- Recovery checks: `Integrated_dataset/tables/gdT_prediction/gdtai_v4_2_recovery_preflight/recovery_checks.csv`
- Approval template:
`Integrated_dataset/logs/gdT_prediction/gdtai_v4_2_recovery_preflight/RECOVERY_EXECUTION_APPROVAL_TEMPLATE.json`
- HTML: `gdT_prediction/gdtai_v4_2_recovery_preflight/index.html`
- PDF: `gdT_prediction/gdtai_v4_2_recovery_preflight/gdtai_v4_2_recovery_preflight_report.pdf`